

Dhinesh Sadhu Subramaniam Ponnarasan

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EDUCATION

State University of New York at Binghamton Clark Fellowship <i>Master of Science - Information Systems with Applied Data Science; Cumulative GPA: 3.21/4.00</i>	Binghamton, NY Aug 2024 – Present
• Coursework: Machine Learning, Generative AI, Python, JavaScript, API Development, Natural Language Processing, Deep Learning.	
Vellore Institute of Technology <i>Post Graduate Program - Data Science; Cumulative GPA: 3.36/4.00</i>	Bangalore, India Aug 2023 – June 2024
• Coursework: Data Structures and Algorithms, Big Data, Artificial Intelligence, C/C++ Programming, Database, Data visualization	
Sri Krishna Arts and Science College <i>Bachelor of Computer Applications; Cumulative GPA: 3.20/4.00</i>	Coimbatore, India Apr 2019 – Jun 2022

EXPERIENCE

AI/ML & Applications Development Intern <i>Uplifty AI</i>	Aug 2025 – Present Austin, Texas, United States
• Developed and implemented AI solutions leveraging Large Language Models (LLMs) and prompt engineering using Python , integrating AI capabilities into production systems that improved recommendation accuracy by 42% through systematic prompt optimization and model fine-tuning techniques.	
• Built automation scripts and tools using Python and Bash for infrastructure management and workflow optimization, implementing CI/CD pipelines and DevOps practices that processed 500K+ records daily, reducing manual operations by 65% through intelligent automation frameworks.	
• Architected AI-driven solutions using LLMs , NLP , and prompt engineering with Python , TensorFlow , and LangChain , collaborating with cross-functional teams to modernize workflows through AI-driven insights, achieving 88% accuracy in content classification tasks.	
• Participated in proof-of-concept projects demonstrating value of AI tools in production environments, utilizing version control systems (Git) , Agile development methodology , and iterative development processes to deliver AI-powered features with measurable business impact.	
Software Development Engineer <i>V3Techserv</i>	June 2022 – July 2023 Chennai, India
• Developed AI solutions and automation tools using Python and Bash scripting , implementing DevOps practices and CI/CD pipelines for infrastructure automation that improved deployment efficiency by 34% through intelligent workflow orchestration and continuous integration.	
• Built enterprise-scale automation scripts with Python and Bash for data pipeline management, processing 2M+ records monthly through automated ETL workflows, utilizing version control (Git) and collaborative development practices to ensure code quality and maintainability.	
• Collaborated with cross-functional teams in Agile development environment to architect solutions for complex infrastructure challenges, leveraging AI-driven insights and automation to modernize legacy systems and enhance operational efficiency across enterprise platforms.	
Software Developer <i>Freelancer</i>	Nov 2021 – May 2022 Coimbatore, India
• Implemented AI-driven automation using Python and Bash scripts for infrastructure operations, developing intelligent pricing and recommendation systems that increased conversion by 28% through LLM integration and prompt engineering techniques.	
• Built automation tools using Python , Bash , and SQL for system management and workflow optimization, implementing DevOps practices and version control with Git to streamline deployment processes and ensure code reliability.	
• Worked with AI tools and prompt engineering to enhance application functionality, collaborating across teams using Agile methodology to deliver iterative improvements, demonstrating adaptability and continuous learning in fast-paced development environment.	

PROJECTS

Enterprise AI Integration Platform with LLM Automation <i>Python, Bash, LLMs, LangChain, Prompt Engineering, Git, CI/CD, DevOps, Automation</i>	Jan 2024 – Mar 2024
• Architected AI solution integrating Large Language Models (LLMs) into enterprise workflows using Python , LangChain , and prompt engineering , processing 1M+ records with intelligent automation that improved operational efficiency by 94% through AI-driven insights and workflow modernization.	
• Developed automation scripts and tools using Python and Bash for infrastructure management, implementing CI/CD pipelines , DevOps practices , and version control (Git) to streamline deployment processes, achieving 89% reduction in manual intervention through intelligent orchestration.	
• Participated in proof-of-concept project demonstrating value of AI tools in enterprise systems, collaborating with cross-functional teams using Agile methodology to deliver AI capabilities, creating comprehensive documentation and implementing monitoring systems for production deployment.	
AI-Driven Infrastructure Automation & LLM Integration System <i>Python, Bash, LLMs, Prompt Engineering, NLP, Git, CI/CD, DevOps, Agile, Automation</i>	May 2023 – Jul 2023
• Developed AI solutions leveraging LLM usage and prompt engineering using Python , TensorFlow , and NLP frameworks , achieving 85% accuracy through systematic prompt optimization and model evaluation, demonstrating extensive background working with AI tools for complex development environments.	
• Built enterprise automation infrastructure using Python and Bash scripting , implementing DevOps practices , CI/CD pipelines , and version control (Git) to automate 500K+ document processing workflows, reducing operational overhead by 60% through intelligent automation and orchestration.	
• Collaborated with teams using Agile development methodology to architect AI-driven solutions , integrating AI capabilities into production systems through proof-of-concept projects , demonstrating continuous learning with cutting-edge tools and modernizing workflows through AI-powered insights.	

TECHNICAL SKILLS

AI & LLM Technologies: Large Language Models (LLMs), Prompt Engineering, LLM Usage, AI Tools, LangChain, AI-Driven Solutions, AI Capabilities Integration, Proof-of-Concept Development
Programming & Scripting: Python, Bash, Shell Scripting, Automation Scripts, Infrastructure Automation, Workflow Automation
DevOps & Infrastructure: DevOps Practices, CI/CD Pipelines, Infrastructure Management, Continuous Integration, Continuous Deployment, Workflow Orchestration, Enterprise Systems
Version Control & Collaboration: Git, GitHub, Version Control Systems, Code Review, Collaborative Development, Agile Development Methodology
AI Development Tools: Fast MCP, MCP Servers, MCP Generation, LangFuse, AI-Based Test Driven Development, Vibe Coding, Cutting-Edge Tools
Machine Learning & NLP: TensorFlow, Keras, Scikit-learn, PyTorch, Natural Language Processing, NLTK, spaCy, Model Fine-Tuning, AI Model Development
Enterprise & Cloud: Enterprise Infrastructure, Cloud Computing, Hybrid Cloud, Mainframe Integration, Complex Development Environments, System Modernization
Data & Analytics: Python Libraries (Pandas, NumPy), SQL, PostgreSQL, Data Processing, ETL Pipelines, Data Analysis, Apache Spark
Development Practices: Agile Methodology, Cross-Functional Collaboration, Proof-of-Concept Projects, Iterative Development, Code Quality, Documentation
Additional Tools: Docker, Jupyter, Tableau, Power BI, Excel, Data Visualization, API Development, REST APIs

PUBLICATIONS

Unsupervised Anomaly Detection for Network Intrusion Using Deep Autoencoders (ICNGCS 2025, IEEE)
• Developed AI solution using deep learning and Python , implementing automation workflows and model deployment achieving 91% accuracy, demonstrating expertise in architecting AI-driven systems for enterprise infrastructure security.
Collaboration Search with Knowledge Sharing and Summarization (ICSES 2024, IEEE)
• Architected AI-powered system leveraging LLM capabilities and NLP , building automation tools with Python for enterprise workflows, demonstrating cross-functional collaboration and AI integration expertise.
Image to Audio Conversion to Aid Visually Impaired People by CNN (ICESC 2023, IEEE)
• Implemented AI solution using deep learning and Python automation , participating in proof-of-concept project that demonstrated AI tool value in real-world applications with comprehensive deployment and testing.